LESSON PLAN

HELICOPTER THEORY (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	14
Total No. of Lesson Plans	27	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Drag analysis, induced, parasite, form, skin friction, profile and total drag. D-V curve.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Drag analysis, induced, parasite, form, skin friction, profile and total drag. D-V curve.
5. Specific Objectives	11.1.2.1 Trainee will be able to describe various types of aerofoil sections, conditions of performance, theories, and concept related with performance

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of D-V curve characteristics; Instructor supervision required for theoretical verification.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) Define the term 'drag' in aerodynamic context.
(b) List the two primary categories of drag.
(c) State the relationship between velocity and parasite drag.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Parasite Drag • Components: Form and Skin Friction • Relationship with Velocity • Minimizing Parasite Drag	L & Ds	PPT, IB	15 min
(b)	• Definition of Induced Drag • Relation to Lift and Angle of Attack • Wingtip Vortices Explanation • Trade-off with Parasite Drag	L & Ds	PPT, PDF	20 min
(c)	• Total Drag Composition • Drag Polar Characteristics • D-V Curve Analysis • Optimum Velocity Concept	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how induced drag changes as airspeed increases.
(b) Differentiate between form drag and skin friction drag.
(c) Identify the point of minimum total drag on the D-V curve.

12. Home Assignment and Follow Up	Prepare a brief note on how wing design modifications can reduce induced drag for the next lesson on Lift augmentation.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Lift augmentation. High lift devices. Boundary Layer control	Refer to the master syllabus reference list	LP 15

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min